

PMLJ'25

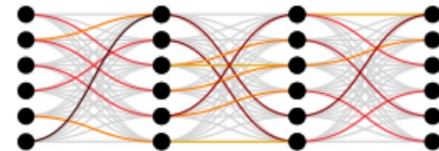
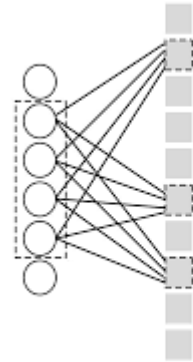
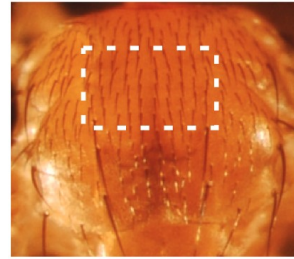
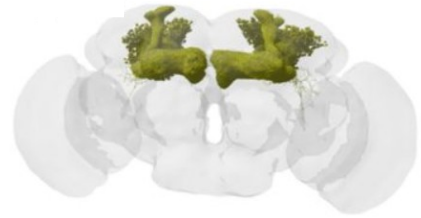
Introduction to Hopfield Networks

Emanuele Natale

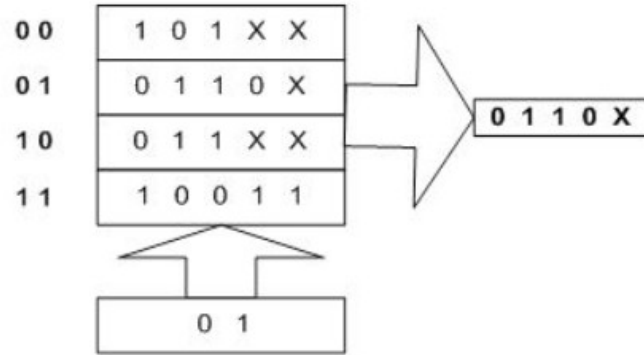
Università di Roma Tor Vergata
21 March 2025

Examples of Algorithms in the Brain

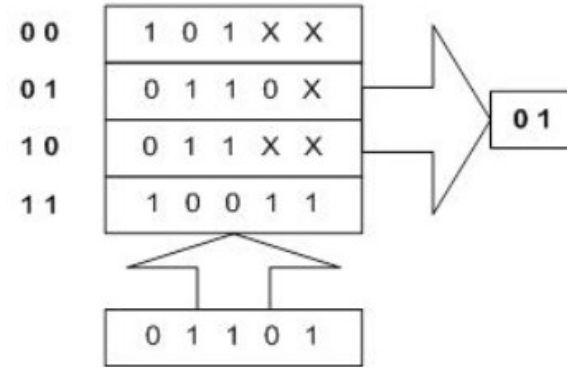
- **Neural data structure** for **novelty detection** by **Navlakha** et al. [PNAS 2018], among others
- **Maximal Independent Set** in the fly nervous system by Afek et al. [Science 2011]
- **Hopfield networks** and **expander codes** by Chaudhuri et al. [PNAS 2019]
- **Synaptic Pruning** and the **Strong Lottery Ticket Hypothesis**



Computer Memory vs Brain Memory



Traditional Memory



**Content Addressable
Memory**

Read operation in traditional memory:

- Input is address location of the content that we are interested in it.
- Output is the content of that address.

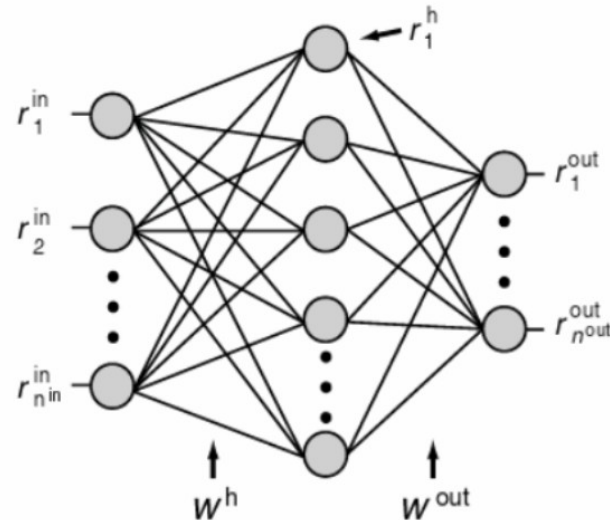
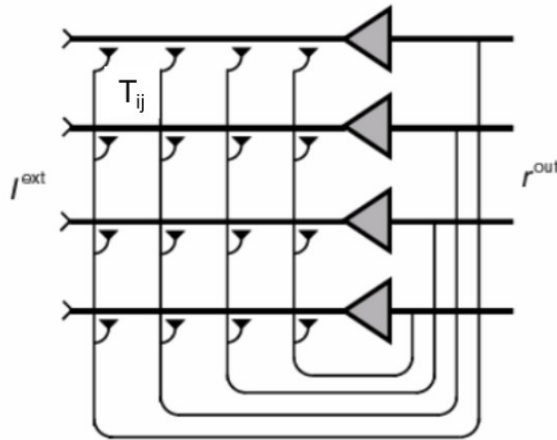
Read op. in CAM memory:

- Input is associated with something stored in the memory.
- Output is location where the associated content is stored.

Feedforward versus recurrent neural networks

Some (incomplete) history

- Rosenblatt (1958): Perceptron
- Steinbuch (1961): Lernmatrix
- Minsky & Papert (1969): XOR
- Willwacher (1976): Thought pump
- Palm (1980)
- Braitenberg (1984)
- Error backpropagation
- Support-vector machines
- Deep learning...



Modern Hopfield Networks

A new frontier for Hopfield networks

[Dmitry Krotov](#) 

[Nature Reviews Physics](#) 5, 366–367 (2023) |

[← Go to ICLR 2021 Conference homepage](#)

Hopfield Networks is All You Need

Hubert Ramsauer, Bernhard Schögl, Johannes Lehner, Philipp Seidl, Michael Kreil, Michael K Kopp, Günter Klambauer, Johannes Brandstetter, Sepp

Published: 12 Jan 2021, Last Modified: 22 Oct 2023 ICLR 2021 Poster Readers:  Everyone 

PNAS

RESEARCH ARTICLE

Building transformers from neurons and astrocytes

Leo Kozachkov^{a,b,1} , Ksenia V. Kastanenka^c, and Dmitry Krotov^{a,1}

Edited by Terrence Sejnowski, Salk Institute for Biological Studies, La Jolla, CA; received November 9, 2022; accepted June 22, 2023

New York Times 1958

NEW NAVY DEVICE LEARNS BY DOING

Psychologist Shows Embryo
of Computer Designed to
Read and Grow Wiser

WASHINGTON, July 7 (UPI)

—The Navy revealed the embryo of an electronic computer today that it expects will be able to walk, talk, see, write, reproduce itself and be conscious of its existence.

The embryo—the Weather Bureau's \$2,000,000 "704" computer—learned to differentiate between right and left after fifty attempts in the Navy's demonstration for newsmen.

The service said it would use this principle to build the first of its Perceptron thinking machines that will be able to read and write. It is expected to be finished in about a year at a cost of \$100,000.

Dr. Frank Rosenblatt, designer of the Perceptron, conducted the demonstration. He said the machine would be the first device to think as the human brain. As do human be-

ings, Perceptron will make mistakes at first, but will grow wiser as it gains experience, he said.

Dr. Rosenblatt, a research psychologist at the Cornell Aeronautical Laboratory, Buffalo, said Perceptrons might be fired to the planets as mechanical space explorers.

Without Human Controls

The Navy said the perceptron would be the first non-living mechanism "capable of receiving, recognizing and identifying its surroundings without any human training or control."

The "brain" is designed to remember images and information it has perceived itself. Ordinary computers remember only what is fed into them on punch cards or magnetic tape.

Later Perceptrons will be able to recognize people and call out their names and instantly translate speech in one language to speech or writing in another language, it was predicted.

Mr. Rosenblatt said in principle it would be possible to build brains that could reproduce themselves on an assembly line and which would be conscious of their existence.

In today's demonstration, the "704" was fed two cards, one with squares marked on the left side and the other with squares on the right side.

Learns by Doing

In the first fifty trials, the machine made no distinction between them. It then started registering a "Q" for the left squares and "O" for the right squares.

Dr. Rosenblatt said he could explain why the machine learned only in highly technical terms. But he said the computer had undergone a "self-induced change in the wiring diagram."

The first Perceptron will have about 1,000 electronic "association cells" receiving electrical impulses from an eye-like scanning device with 400 photo-cells. The human brain has 10,000,000,000 responsive cells, including 100,000,000 connections with the eyes.

Hopfield Networks: the Model

A model of **content-addressable associative memory**

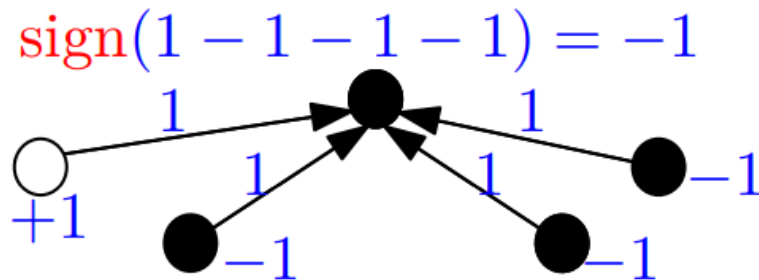
N-node **weighted** graph G

Each node v has binary state $s_v \in \{-1, +1\}$ (McCulloch & Pitts, 1943)

Dynamics: Pick a node v at random and set

$$s_v \leftarrow \text{sign}\left(\sum_u s_u w_{u,v}\right) \quad (\text{local \& deterministic})$$

until changes don't occur anymore

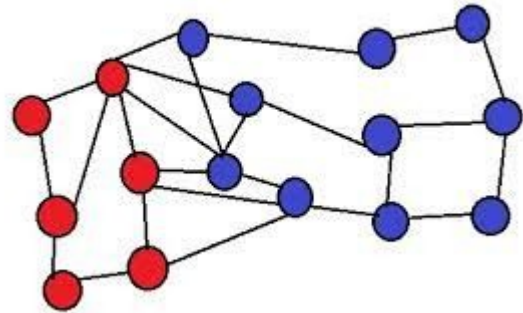


Convergence (Parallel Case)

At the beginning every node is either **blue** or **red**.

At each round, each node looks at the neighbors (assume **odd degree**) and adopt the opinion of the majority.

Prove that the system is eventually **at most two periodic**.



Asynchronous case:

Convergence to binary **N**-dimensional vectors $\{v^{(i)}\}_i$

Hebbian Learning

Energy function: $H = -\frac{1}{2} \sum_{i \neq j} w_{ij} s_i s_j$

How to set **weights** $w_{u,v}$?

Hebbian learning:

$$w_{ij} = \frac{1}{N} \sum_{k=1}^p v_i^{(k)} v_j^{(k)}$$

activity of neuron
k in pattern **i**



“fire together, wire together”

[Hopfield, Proc. Natl. Acad. Sci. 1984]

Neurons with graded response have collective computational properties like those of two-state neurons

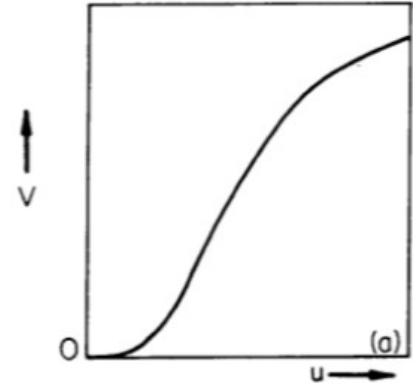
$$C_i(du_i/dt) = \sum_j T_{ij}V_j - u_i/R_i + I_i$$

$$V_i = g_i(u_i) \quad \text{high dimensional} \\ \text{\& nonlinear dynamics}$$

$u(t)$: internal activity - e.g. somatic potential

$V(t)$: neural output - e.g. firing rate

T_{ij} : synaptic couplings (synaptic symmetry $T_{ij} = T_{ji}$)



Characteristics of time-dependent solutions?

Stationary solutions (and their stability)?

Computation?

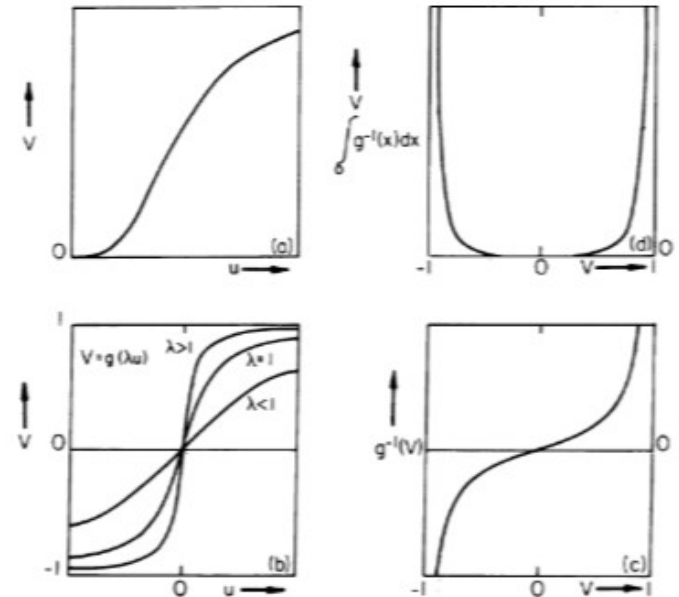
Energy Function

$$E = -\frac{1}{2} \sum_{i,j} T_{ij} V_i V_j + \sum_i (1/R_i) \int_0^{V_i} g_i^{-1}(V) dV + \sum_i I_i V_i.$$

Energy (Lyapunov) function

E is bounded below and non-increasing along every solution.
 $\implies$ system has to approach (local) minima.

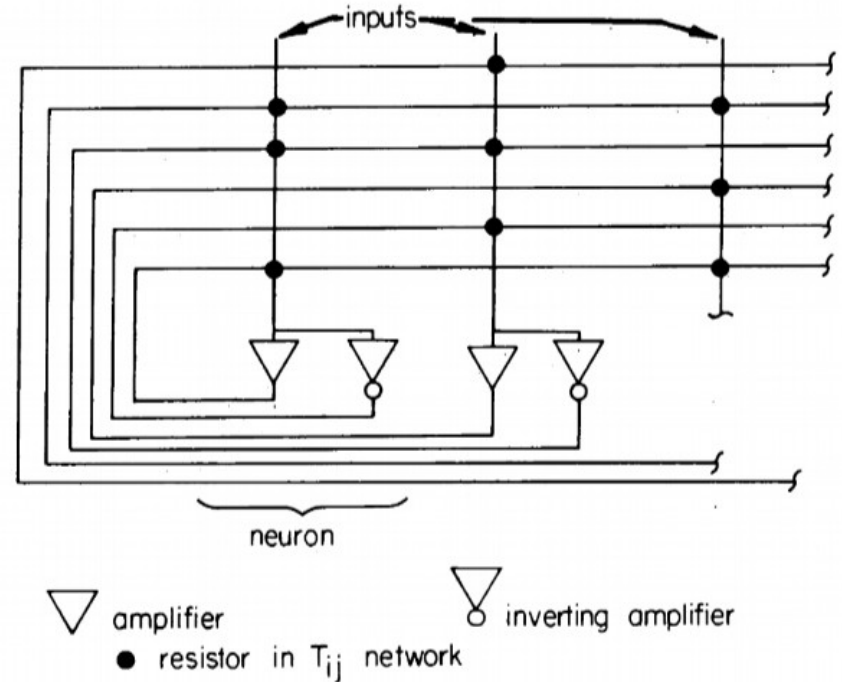
Here: stationary solutions
 $\implies$ No explicit solutions have to be calculated!



Electrical analog

$$C_i(du_i/dt) = \sum_j T_{ij}V_j - u_i/R_i + I_i$$
$$V_i = g_i(u_i)$$

[Hopfield PNAS 1984]

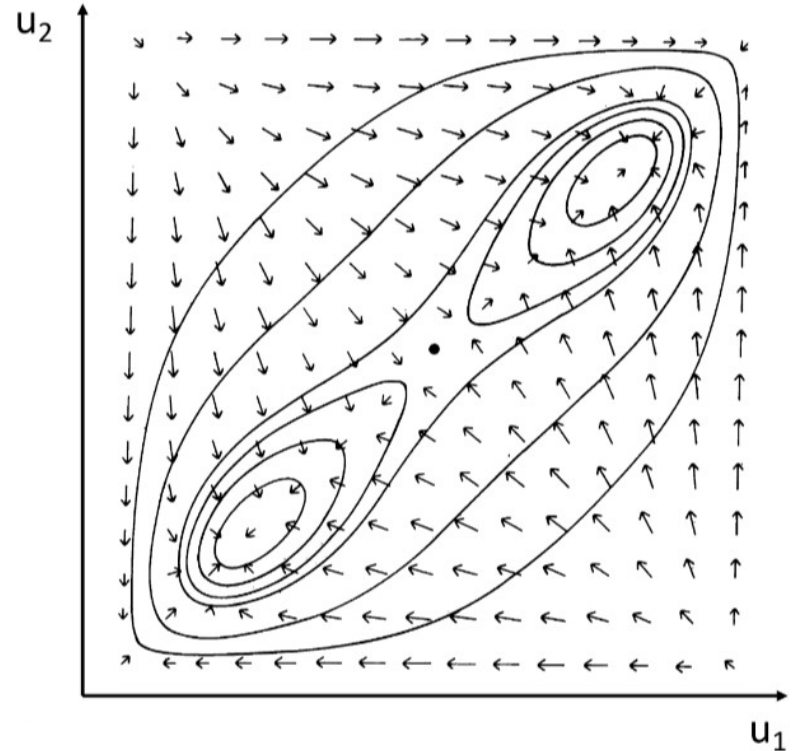


Easy to [implement](#)

Energy Landscape

System with 2 neurons and 2
point attractors

Computation done by the
physics of the system (and not
by digital device, storing info by
location)



Some variants of the Hopfield Model

- Diluted networks
- Clipped synapses
- Binary synapses
- Parallel updating
- Parallel updating and delays
- Block-sequential updating
- Correlated patterns (incl. hierarchies)
- Bias by external input
- ...
- and many more

[Parisi, J. Phys. A 1986]

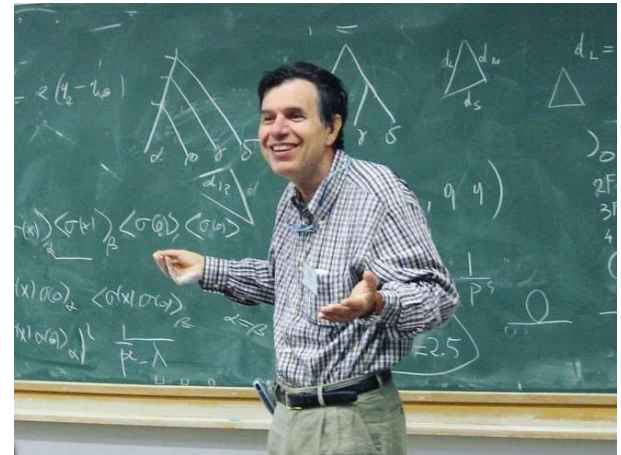
A memory which forgets

J. Phys. A: Math. Gen. **19** (1986) L617-L620. Printed in Great Britain

$$w_{ij}^{new} = f(w_{ij} + C v_i^{(p+1)} v_j^{(p+1)})$$

Abstract. The model of Hopfield for a neural network with associative memory is modified by the introduction of a **maximum value for the synaptic strength**; in this way **old patterns are automatically forgotten and the memory recalls only the most recent ones**. If the parameters are correctly chosen, the memory never goes into the state of total confusion characteristic of the Hopfield model.

$$\begin{aligned} f(x) &= -A && \text{for } x < -A \\ f(x) &= x && \text{for } -A < x < A \\ f(x) &= A && \text{for } A < x. \end{aligned}$$



Attractor networks: summary & outlook

Point attractors.

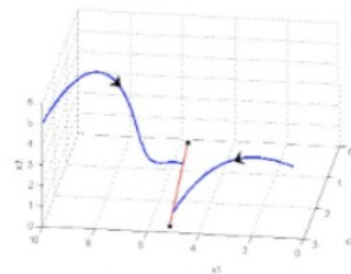
- Associative memory

Continuous attractors.

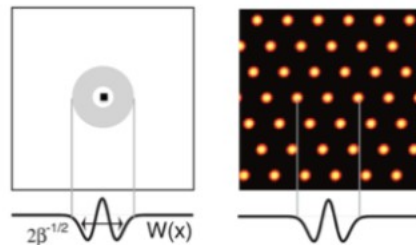
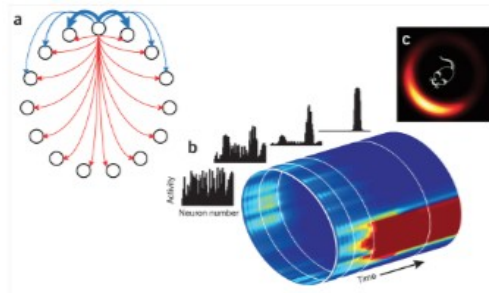
- 1D Line attractors.
 - Ring attractors
 - orientation tuning in visual cortex
 - head direction system ...
- 2D toroidal attractors
 - grid cells in entorhinal cortex

Time dependent attractors.

- Pattern sequences (sensory, motor)



Line Attractor



Absolute Stability of Global Pattern Formation and Parallel Memory Storage by Competitive Neural Networks

[Cohen & Grossberg IEEE 1982]

≈3k cit.s

MICHAEL A. COHEN AND STEPHEN GROSSBERG

Stigler's
law of
eponymy

Abstract—The process whereby input patterns are transformed and stored by competitive cellular networks is considered. This process arises in such diverse subjects as the short-term storage of visual or language patterns by neural networks, pattern formation due to the firing of morphogenetic gradients in developmental biology, control of choice behavior during macromolecular evolution, and the design of stable context-sensitive parallel processors. In addition to systems capable of approaching one of perhaps infinitely many equilibrium points in response to arbitrary input patterns and initial data, one finds in these subjects a wide variety of other behaviors, notably traveling waves, standing waves, resonance, and chaos. The question of what general dynamical constraints cause global approach to equilibria rather than large amplitude waves is therefore of considerable interest. In another terminology, this is the question of whether global pattern formation occurs. A related question is whether the global pattern formation property persists when system parameters slowly change in an unpredictable fashion due to self-organization (development, learning). This is the question of absolute stability of global pattern formation. It is shown that many model systems which exhibit the absolute stability property can be written in the form

$$\frac{dx_i}{dt} = a_i(x_i) \left[b_i(x_i) - \sum_{k=1}^n c_{ik} d_k(x_k) \right] \quad (1)$$

$i = 1, 2, \dots, n$, where the matrix $C = \|c_{ik}\|$ is symmetric and the system as a whole is competitive. Under these circumstances, this system defines a

global Liapunov function. The absolute stability of systems with infinite but totally disconnected sets of equilibrium points can then be studied using the LaSalle invariance principle, the theory of several complex variables, and Sard's theorem. The symmetry of matrix C is important since competitive systems of the form (1) exist wherein C is arbitrarily close to a symmetric matrix but almost all trajectories persistently oscillate, as in the voting paradox. Slowing down the competitive feedback without violating symmetry, as in the systems

$$\begin{aligned} \frac{dx_i}{dt} &= a_i(x_i) \left[b_i(x_i) - \sum_{k=1}^n c_{ik} d_k(x_k) \right] \\ \frac{dy_i}{dt} &= e_i(x_i) [f_i(x_i) - y_i], \end{aligned}$$

also enables sustained oscillations to occur. Our results thus show that the use of fast symmetric competitive feedback is a robust design constraint for guaranteeing absolute stability of global pattern formation.

I. INTRODUCTION: ABSOLUTE STABILITY OF GLOBAL PATTERN FORMATION IN SELF-ORGANIZING NETWORKS

THIS ARTICLE proves a global limit theorem for a class of n -dimensional competitive dynamical systems that can be written in the form

$$\dot{x}_i = a_i(x_i) \left[b_i(x_i) - \sum_{k=1}^n c_{ik} d_k(x_k) \right], \quad (1)$$

$i = 1, 2, \dots, n$, where the coefficients $\|c_{ij}\|$ form a symmet-

Manuscript received August 1, 1982; revised April 4, 1983. This work was supported in part by the National Science Foundation under Grant NSF IST-80-00257 and in part by the Air Force Office of Scientific Research under Grant AFOSR 82-0148.

The authors are with the Center for Adaptive Systems, Department of Mathematics, Boston University, Boston, MA 02215.

[Hopfield 1982]

≈27k cit.s

Proc. Natl. Acad. Sci. USA
Vol. 79, pp. 2554–2558, April 1982
Biophysics

Neural networks and physical systems with emergent collective computational abilities

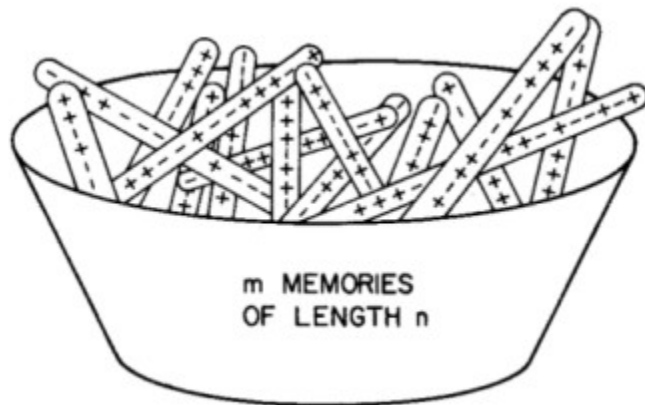
(associative memory/parallel processing/categorization/content-addressable memory/fail-soft devices)

Simpler
is
Better

ABSTRACT Computational properties of use to biological organisms or to the construction of computers can emerge as collective properties of systems having a large number of simple equivalent components (or neurons). The physical meaning of content-addressable memory is described by an appropriate phase space flow of the state of a system. A model of such a system is given, based on aspects of neurobiology but readily adapted to integrated circuits. The collective properties of this model produce a content-addressable memory which correctly yields an entire memory from any subpart of sufficient size. The algorithm for the time evolution of the state of the system is based on asynchronous parallel processing. Additional emergent collective properties include some capacity for generalization, familiarity recognition, categorization, error correction, and time sequence retention. The collective properties are only weakly sensitive to details of the modeling or the failure of individual devices.

Capacity of Hopfield Networks

How many vectors before **errors** appear?



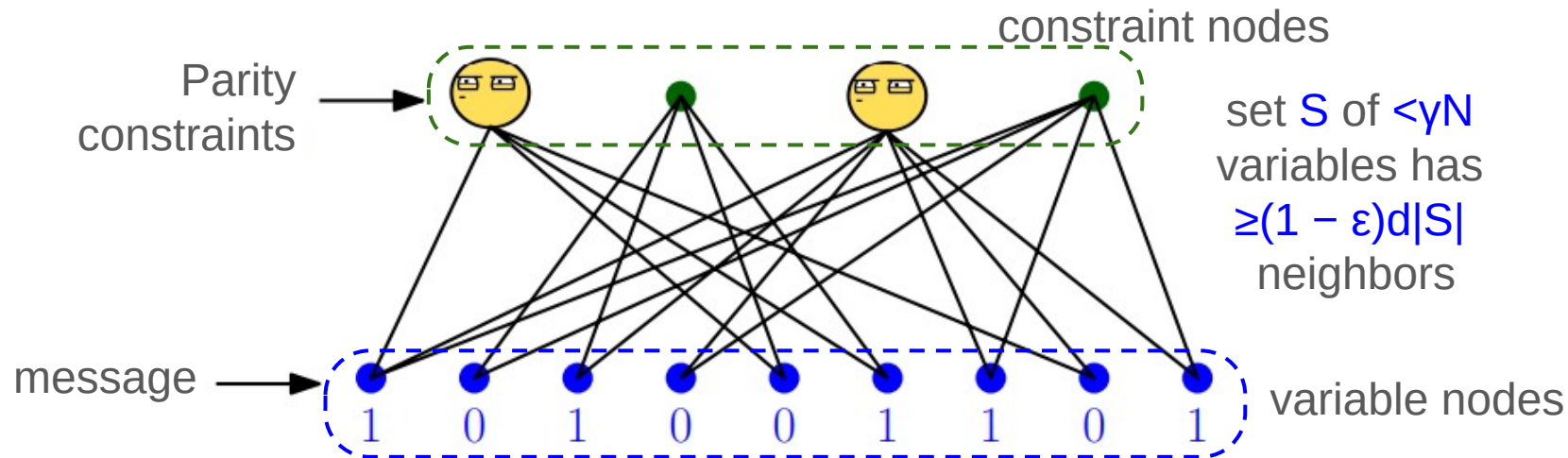
For **random vectors**, capacity is $\Theta(N)$

For **structured patterns** with other dynamics, capacities are $\approx N, 2^{\sqrt{n}}, 2^{\frac{n}{\log n}}$
(but **not robust**)

Problem. **Exponential capacity** $2^{\Omega(n)}$ **in Hopfield networks** with structured patterns?

From Expander Codes to Hopfield Networks

Expander Codes. [Sipser & Spielman '96]



Dynamics. Flip a variable with $\# \text{unsatisfied} > \text{satisfied neighbors}$

Thm. Dynamics recovers $2^{\Omega(n)}$ vectors, if $< \frac{\gamma}{2}N$ entries are flipped.

Exponential-Capacity Hopfield Network [Chauduri&Fiete '19].

Constraint nodes → small Hopfield networks.

Dynamics → pick a random node and flip it to majority

Conclusions

Hopfield networks are a famous model of associative memory.

Techniques from computer science allows to construct efficient Hopfield networks.

